Berkeley Engineering | BerkeleyHaas

PROFESSIONAL CERTIFICATE IN MACHINE LEARNING AND ARTIFICIAL INTELLIGENCE

Office Hour with Mani K

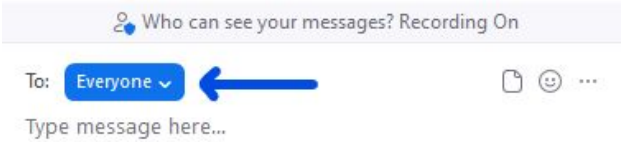
Feb 18, 2026

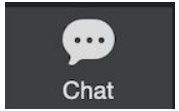
GUIDELINES ON ZOOM

Audio & Video

- Keep your microphone muted while presenters are speaking.
- Keep your camera on, preferably.

Chat or use Q&A





Introduction

Mani Kannappan

Learning Facilitator - section 'B'

- Product Management, Advisor, Entrepreneur
- CPO - Vogo.com, travel-tech startup
- MBA, Berkeley Haas

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General Announcements

Good to have a playground to learn

- Anaconda Installation

  https://www.anaconda.com/download

- Creating Environments (Mac) - Medium Article Link

or Alternatives

- https://www.kaggle.com/
- https://colab.research.google.com/

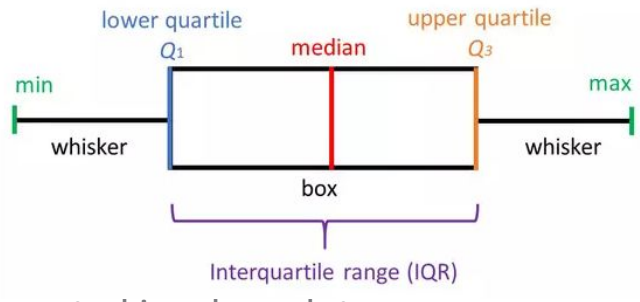
Agenda

- Statistics & Normal Distribution
- Samples & Population
- Jupyter notebook example on Conda env - exercise with Pandas
- Link to Lending club Notebook & data file
  - https://drive.google.com/file/d/1yGGo8t2vPizE30oxX1jd-E-p6qjTIbJK/view?usp=sharing
  - https://drive.google.com/open?id=1iNpJTCbrZ44T3qQFPm7PZZ413OUZi8iz

- First Moment : Mean
- Second moment : SD or variance
- Third moment: Skewness
- Fourth moment : Kurtosis

Typically we use the 5 number summary to describe numerical datasets using the “describe”function in Pandas Library inside the Jupyter notebook

- Max, Min & average
- lower quartile
- median or 2nd quartile
- upper quartile



The 5 number summary can also be conveniently represented in a box plot

- Measure of the spread of the observations from the mean
- Higher variance means larger the spread
- Variance is defined as as the average of the squared deviation from the mean

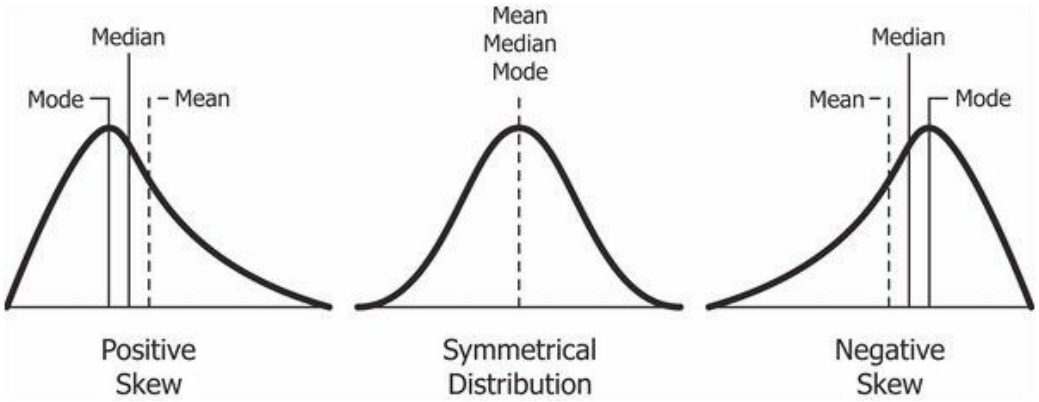
$$s^2 = \frac{(y_1 - \bar{y})^2 + (y_2 - \bar{y})^2 + \cdots + (y_n - \bar{y})^2}{n-1}$$

- Where $\bar{y}$ is the mean and n is the number of observation

	Value	Deviation	Squared
y_1	4	4 − 7 = −3	9
y_2	0	0 − 7 = −7	49
y_3	8	8 − 7 = 1	1
y_4	11	11 − 7 = 4	16
y_5	12	12 − 7 = 5	25
Sum	35	0	100

- $\bar{y}$ = 35/5 = 7 (mean)
- n-1 = 4
- s^2 = 100/4 = 25
- Standard deviation is the square root of the variance
- s = sqrt(s2) = 5

- Third moment: Skewness



- Pandas
  - df.skew()

- Fourth moment :
  - Kurtosis - measure of how the distribution matches a normal distribution
  - Higher the kurtosis, longer the tail
  - https://pandas.pydata.org/pandas-docs/stable/reference/api/pandas.DataFrame.kurtosis.html

```
df.kurtosis()
```

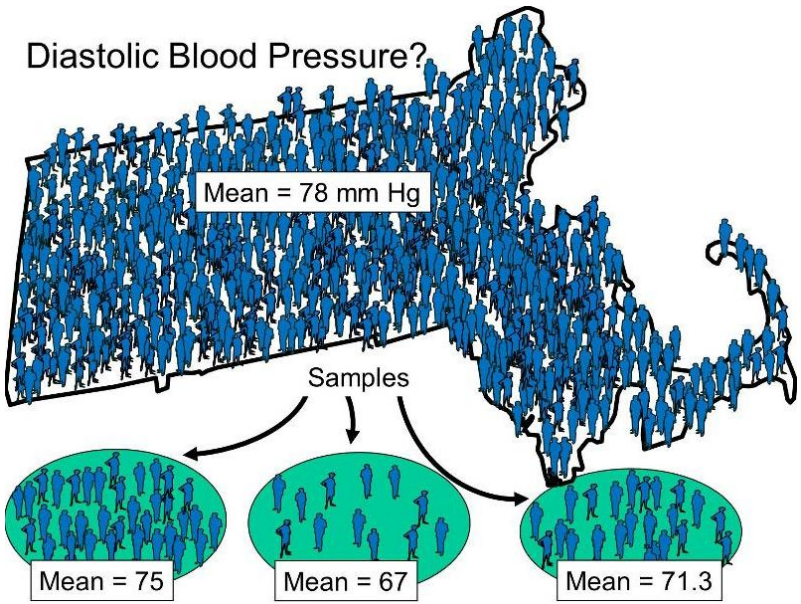
Samples & Population

Population:

- All available observations in the data set
- Hard,expensive and often impossible to get data from the entire population.
    - US census is one example (taken once every 10 years)

Sample:

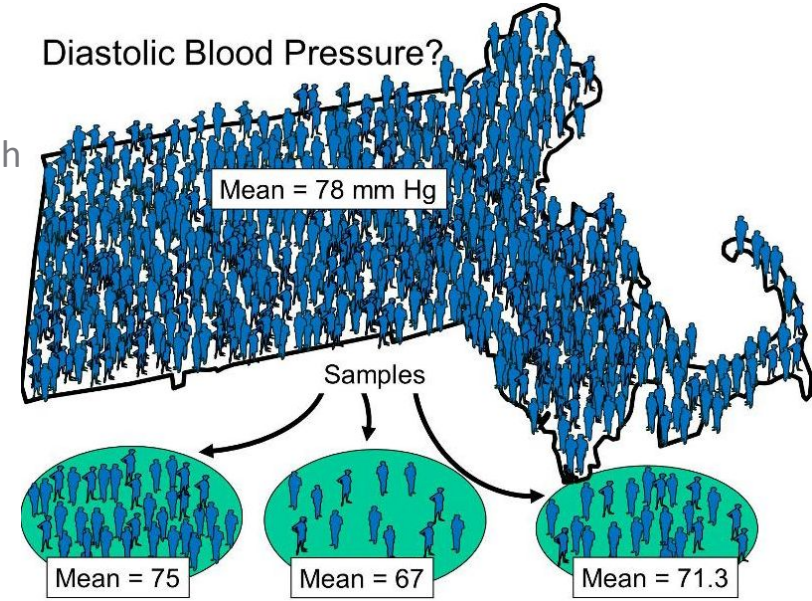
- A subset of random data that represents the population



Samples & Population

For example in,

- If we take different sample sets of the blood pressure each with varying sample sizes
- The first sample set may have an average of say 75
- The second sample set may have an average of 67
- The third sample may have an average of 71.3.. so on and so forth
- And if we plot the mean of each sample sets the plot will be normally distributed

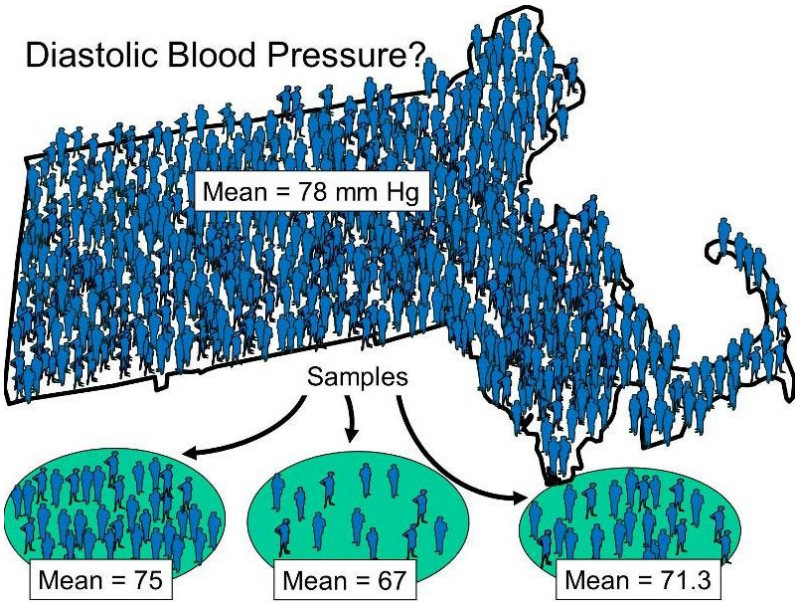


Samples & Population

For example

- if the sample mean = 72
- and the standard deviation (SD) s = 2
- And if all conditions are met (more on that later)

- then we can infer (approximately) with 95% confidence that the BP is between
    - 72 + 2*SD = 72 + 4 = 76
    - 72 - 2 *SD = 72 -4 = 68
- With 95% confidence the average blood pressure will be between 68 and 76 for the entire population



Pandas Common Used

Data Loading

pd.read_csv('file.csv')

pd.read_excel('file.xlsx')

Inspection

df.info() & df.shape

df.describe()

df.head()

df.tail()

Selection

```
df['column']

df[['col1','col2']]

df.loc[row_selector, col_selector]

df.iloc[row_index, col_index]
```

Filtering

```
df[df['column'] > value]

df.query('column > value')
```

Column Operations

```
df['new'] = df['A'] + df['B']
df['col'].sum()
df['col'].mean()
df['col'].std()
```

Handling Missing Values

```
df.isnull().sum()
df.dropna()
df.fillna(value)
```

Visualization

```
df.plot(kind='hist')

df.plot(kind='scatter', x='col1', y='col2')

df.plot(kind='line')
```